

PUBLIC TRANSIT PLANNER

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Urban commuters often struggle with fragmented transit information — scattered bus and metro schedules, unpredictable delays, and no unified way to compare routes across different modes of transport. This project proposes a Public Transit Planner, a digital platform that helps users plan, track, and optimize their daily commute across buses, metros, and other public transit options. The system integrates real-time vehicle tracking, route optimization, and fare estimation into a single interface, allowing users to input a source and destination and receive the fastest, cheapest, or least-transfer route instantly. Key features include live arrival predictions, multi-modal route comparison, offline access to schedules, and accessibility-aware routing for differently abled commuters. By consolidating transit data that is currently spread across multiple apps and physical timetables, the platform reduces commute uncertainty, encourages public transport adoption over private vehicles, and supports more sustainable urban mobility. The proposed system is designed to be scalable across cities with varying transit infrastructure.

Signature of the Guide